



The magnitude for Nakayama algebras

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ABSTRACT

The magnitude for algebras is a generalization of the Euler characteristic. We investigate the magnitude for Nakayama algebras. Using Ringel's resolution quiver, the existence and the value of rational magnitude is given. As a result, we show directly that the finite global dimension criteria for Nakayama algebras of Madsen and the first author are equivalent.

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1. Introduction

Let R be a commutative ring. Given a positive integer n , we write $u \in R^n$ for the column vector with $u_i = 1$ for all $1 \leq i \leq n$. Let C be an $n \times n$ matrix in R . A weighting on C is a column vector α such that $C\alpha = u$. A coweighting on C is a row vector β such that $\beta C = u^*$, where u^* is the transpose of u .

Following [4], a matrix C is said to have magnitude if it admits at least one weighting α and at least one coweighting β . The magnitude of C is

$$m(C) = \sum_i \alpha_i = \sum_i \beta_i.$$

A matrix has magnitude if and only if its transpose has magnitude. If a matrix C has magnitude, the magnitude of C is independent of the weighting and the coweighting chosen. If C is an invertible matrix, then the weighting and coweighting are unique and the magnitude of C is the sum of all entries in the inverse matrix of C ; see [1,4].

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Let A be a finite dimensional algebra over a field k . If A has finite global dimension, then the Cartan matrix C_A of A is invertible in the ring of integers. Following [1], the magnitude of C_A is equal to the Euler characteristic of A . If A has infinite global dimension, the Euler characteristic is an infinite sum while the magnitude may exist in the field of rational numbers.

In the present paper, we study the magnitude of the Cartan matrix for any Nakayama algebra. Using Ringel's resolution quiver, we prove the existence and give the value of magnitude in the field of rational numbers. We describe syzygies of simple modules and give a special coweighting. As a result, we show directly the following two finite global dimension criteria for Nakayama algebras are equivalent. In the last section, we compute the Cartan determinant of any graded Nakayama algebra.

Theorem 1.1. [5] *A Nakayama algebra A has finite global dimension if and only if there exists a simple A -module of even injective dimension.*

Theorem 1.2. [12] *A Nakayama algebra A has finite global dimension if and only if the resolution quiver of A is connected of weight 1.*

Through this paper, k is a fixed field. All modules are finite dimensional right modules.

2. Magnitude of a Nakayama algebra

Let n be a positive integer and Δ_n be the following quiver

$$1 \begin{array}{c} \xrightarrow{x_1} \\ \xleftarrow{x_n} \end{array} 2 \xrightarrow{x_2} \cdots \xrightarrow{x_{n-1}} n.$$

An algebra A over a field k is called a Nakayama algebra of order n if A is isomorphic to the quotient algebra of $k\Delta_n$ by an ideal generated by a finite set $\mathcal{R}$ of nontrivial paths. If $\mathcal{R}$ consists of paths of length ≥ 2 , then A is a cyclic Nakayama algebra. If there is only one arrow in $\mathcal{R}$, then A is a linear Nakayama algebra. We allow more than one arrow in $\mathcal{R}$.

Let A be a Nakayama algebra of order n . For any $1 \leq i \leq n$, let e_a be the idempotent, S_a the simple module, P_a the projective module, and I_a the injective module at a . Define $\tau(a) = a+1$ for any $1 \leq a \leq n-1$ and $\tau(n) = 1$. Denote by p_a the dimension of P_a . Then $p_a \leq p_{\tau(a)} + 1$ for any a . The sequence $(p_1, p_2, \dots, p_n)$ is called an admissible sequence of A .

The resolution quiver R_A is introduced by Ringel in [6]. The vertices are $S_1, S_2, \dots, S_n$ and there is a unique arrow from S to $\gamma(S) = \tau \operatorname{soc} P(S)$ for each simple module S . The resolution quiver of A can be realized as the functional graph of the Gustafson function

$$\gamma(a) = \tau^{p_a}(a)$$

for any $1 \leq a \leq n$; see [2]. Let q_a be the dimension of I_a . The function

$$\psi(a) = \tau^{-q_a}(a)$$

gives rise to the coresolution quiver R_A^* of A .

A vertex in the resolution quiver is called cyclic if it lies on a cycle. Given a cycle I in R_A , the periodicity p of I is the number of vertices in I . The weight w of I is $\sum_a p_a/n$, where a runs through all vertices in I . It turns out that p and w are relatively prime integers. Moreover, p and w are independent of the cycle chosen; see [10].

Let $\mathbf{d}$ be a grading on A , i.e. $\mathbf{d}$ is a map from the arrows to $\mathbb{Z}$. For any integer ℓ , we write A_ℓ for the degree ℓ part of A . The graded Cartan matrix $C_A^{\mathbf{d}}(t)$ is a matrix in the ring $\mathbb{Z}[t^{\pm 1}]$ of Laurent polynomials over $\mathbb{Z}$ and

$$c_{ab}(t) = \sum_{\ell} \dim_k(e_b A_{\ell} e_a) t^{\ell}$$

for all $1 \leq a, b \leq n$. If $\mathbf{d}$ is identically zero, then $C_A^{\mathbf{d}}(t) = C_A$ is the Cartan matrix of A .

Given a subset I of $\{1, 2, \dots, n\}$, we write $\epsilon_I \in \mathbb{Z}^n$ for the characteristic vector of I . Here, $\epsilon_{I,a} = 1$ if $a \in I$ and $\epsilon_{I,a} = 0$ otherwise.

Lemma 2.1. *Let C_A be the Cartan matrix of A .*

- (1) *If I is a cycle in R_A , then ϵ_I/w is a weighting on C_A ;*
- (2) *If α is a weighting on C_A and a is not cyclic, then $\alpha_a = 0$;*
- (3) *If α is a weighting on C_A and a, b lie on the same cycle, then $\alpha_a = \alpha_b$.*

Proof. (1) Denote by $\mathbf{dim} M$ the dimension vector of a module M . Let $a_1, a_2, \dots, a_p$ be the vertices in I such that $a_{i+1} = \gamma(a_i)$ for all $1 \leq i \leq p-1$ and $a_1 = \gamma(a_p)$. Then the composition factors of $P_{a_1}, P_{a_2}, \dots, P_{a_p}$ are consecutive. Since I is a cycle in R_A , each simple module occurs w times. Then we have

$$\mathbf{dim} P_{a_1} + \mathbf{dim} P_{a_2} + \dots + \mathbf{dim} P_{a_p} = w \cdot (1, 1, \dots, 1).$$

It follows that $C_A \epsilon_I = w \cdot u$.

(2&3) Let $I_1, I_2, \dots, I_c$ be all cycles in R_A . Since $I_1, I_2, \dots, I_c$ are pairwise disjoint, the vectors $\epsilon_{I_1}, \epsilon_{I_2}, \dots, \epsilon_{I_c}$ are linearly independent. By [12, 4.1], the number of cycles in R_A is equal to $\text{corank } C_A + 1$. We know that any weighting on C_A is a linear combination of vectors of the form ϵ_I/w , where I is a cycle. Then the statement holds. $\square$

Similarly, one can show that C_A has a coweighting in $\mathbb{Q}$.

Theorem 2.2. *Let A be a Nakayama algebra and C_A be the Cartan matrix of A . Then C_A has magnitude in $\mathbb{Q}$. The magnitude of C_A is*

$$m(C_A) = p/w$$

where p is the periodicity and w is the weight of R_A .

Remark 2.3. Since p, w are relatively prime, it follows that

$$p(R_A) = p(R_A^*), \quad w(R_A) = w(R_A^*).$$

This is proven in [11] by different means; see also [7].

Let $\mathbf{d}$ be a positive grading on A , i.e., $\mathbf{d}(x_a)$ is positive for every arrow x_a . Since $C_A^{\mathbf{d}}(0)$ is the identity matrix, the graded Cartan matrix $C_A^{\mathbf{d}}(t)$ is invertible in the field $\mathbb{Q}(t)$ of rational fractions over $\mathbb{Q}$.

Denote by $\alpha^{\mathbf{d}}(t)$ the unique weighting on $C_A^{\mathbf{d}}(t)$. The a -th entry $\alpha_a^{\mathbf{d}}(t)$ is related to the minimal graded injective resolution of S_a ; see [13, 1.5].

For a graded module M , denote by $M(b)$ the b -th shift of M . Let

$$\xi: 0 \rightarrow S_a \rightarrow I_{c_0}(b_0) \rightarrow I_{c_1}(b_1) \rightarrow \dots$$

be the minimal graded injective resolution of S_a . Then we have

$$\alpha_a^{\mathbf{d}}(t) = \sum_i (-1)^i t^{b_i} \quad (2.1)$$

in the ring $\mathbb{Z}[[t]]$ of power series over $\mathbb{Z}$. For any $i \in \mathbb{N}$ the difference $b_{i+1} - b_i$ is equal to the sum $\sum_e \mathbf{d}(x_e)$, where S_e runs through all composition factors of the i -th cosyzygy of S_a . Since $\mathbf{d}$ is positive, $\{b_i\}$ is strictly increasing.

For a module M , we denote by $\text{id } M$ the injective dimension of M and by $\text{pd } M$ the projective dimension of M .

Proposition 2.4. *Let A be a Nakayama algebra with positive grading $\mathbf{d}$.*

- (1) *If $\text{id } S_a$ is odd, then $\alpha_a^{\mathbf{d}}(1) = 0$;*
- (2) *If $\text{id } S_a$ is even, then $\alpha_a^{\mathbf{d}}(1) = 1$;*
- (3) *If $\text{id } S_a$ is infinite, then $0 < \alpha_a^{\mathbf{d}}(1) < 1$;*
- (4) *If a is not cyclic, then $\alpha_a^{\mathbf{d}}(1) = 0$;*
- (5) *If a, b lie on the same cycle, then $\alpha_a^{\mathbf{d}}(1) = \alpha_b^{\mathbf{d}}(1)$.*

Proof. (1&2) This follows directly from equation (2.1).

(3) Suppose $\text{id } S_a$ is infinite. Since A is of finite representation type, the minimal injective resolution of S_a is periodic after finitely many steps. By (2.1), there are $f(t)$ and $g(t)$ such that

$$\begin{aligned} \alpha_a^{\mathbf{d}}(t) &= f(t) + g(t)/(1 - t^{b_{2r+2p} - b_{2r}}), \\ f(t) &= t^{b_0} - t^{b_1} + \dots + t^{b_{2r-2}} - t^{b_{2r-1}}, \\ g(t) &= t^{b_{2r}} - t^{b_{2r+1}} + \dots + t^{b_{2r+2p-2}} - t^{b_{2r+2p-1}}. \end{aligned}$$

Since $\{b_i\}$ is strictly increasing, then $\alpha_a^{\mathbf{d}}(1)$ is a proper fraction.

(4&5) We know that $\alpha^{\mathbf{d}}(1)$ is a weighting on C_A . Then the statements follow from Lemma 2.1. $\square$

Now we let $\mathbf{d}$ be the length grading, i.e., $\mathbf{d}(x_a) = 1$ for any arrow x_a . Then $\alpha_a(t) = \alpha_a^{\mathbf{d}}(t)$ is determined by the dimension of cosyzygy modules of S_a . If a is not cyclic, then $\alpha_a(1) = 0$. It remains to calculate $\alpha_a(1)$ for any cyclic vertex a in the resolution quiver.

Lemma 2.5. [5, 3.7] *Let a be a vertex in R_A . Then a is cyclic if and only if $\text{id } S_a$ is even or infinite.*

Suppose A has infinite global dimension and a is cyclic in R_A . By Theorem 1.1, $\text{id } S_a$ is infinite. Assume the minimal injective resolution of S_a is periodic after $2r$. We claim that the periodicity is $2p$. Then

$$\alpha_a(1) = \sum_{i=0}^{p-1} \dim_k \Sigma^{2r+2i}(S_a)/nw.$$

Here, Σ^i denotes the i -th cosyzygy of a module. Suppose M is an indecomposable module and M has a periodic minimal injective resolution

$$0 \rightarrow M \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$$

By the dual version of [6, 3.2] $\text{soc } I_0$ and $\text{soc } I_1$ are cyclic in the coresolution quiver R_A^* . Moreover, we have

$$\text{soc } \Sigma^{2p} M \cong \psi^p(\text{soc } M) \cong \text{soc } M.$$

Since

$$\sum_{i=0}^{p-1} \dim_k I^{2p+1} = \sum_{i=0}^{p-1} \dim_k I^{2p} = nw,$$

then $\dim_k M = \dim_k \Sigma^{2p} M$. It follows that $\Sigma^{2p} M \cong M$.

Let n_a be the cardinality of the component containing a in R_A .

Theorem 2.6. *Let A be a Nakayama algebra with length grading.*

- (1) *If a is not cyclic in R_A , then $\alpha_a(1) = 0$.*
- (2) *If a is cyclic in R_A , then $\alpha_a(1) = n_a/nw$.*

Suppose A has finite global dimension and a is cyclic. By Theorem 1.2 $w = 1$ and $n_a = n$. By Lemma 2.5 $\text{id } S_a$ is even and $\alpha_a(1) = 1$. So Theorem 2.6 holds in this case. We study syzygy modules of simple modules and prove Theorem 2.6 for A with infinite global dimension in the next section.

Remark 2.7. Using Theorem 2.6, one can prove the equivalence of Theorem 1.1 and Theorem 1.2 directly.

3. Syzygy modules of a simple module

Let A be a Nakayama algebra. A vertex a in R_A is called a leaf if there is no arrow ending at a . By [12, 3.3], a is a leaf in R_A if and only if $\text{id } S_a = 1$. The algebra A is selfinjective if and only if there is no leaf in R_A .

Let $e = \sum_{a \in \text{Im } \gamma} e_a$ be the sum of the idempotent at all non-leaf vertices and let $\epsilon(A) = eAe$. Then $\epsilon(A)$ is a Nakayama algebra with fewer vertices. Moreover, $\epsilon(A) = A$ if and only if A is selfinjective. If A is a cyclic Nakayama algebra, then $\epsilon(A)$ coincides with Sen's ϵ -construction [8,9]. By [3, 3.8], the resolution quiver of $\epsilon(A)$ equals R_A with all leaves removed.

Let $\text{mod } A$ be the category of finite dimensional modules over A . By [3, 3.6], there is an adjoint pair of exact functors

$$\text{mod } A \begin{matrix} \xleftarrow{G=(-) \otimes_{eAe} eA} \\ \xrightarrow{F=(-)e} \end{matrix} \text{mod } \epsilon(A).$$

Denote by $\mathcal{F}$ the image of $\text{mod } \epsilon(A)$ in $\text{mod } A$ and by $E(S_a)$ the image of S_a for each a in $\text{Im } \gamma$. Then $\mathcal{F}$ is an exact subcategory of $\text{mod } A$ which is closed under projective covers, and $\{E(S_a) \mid a \in \text{Im } \gamma\}$ is a complete set of simple objects in $\mathcal{F}$.

We need the following; see [7, B.2].

- (1) $\text{top } E(S_a) \cong S_a$ for any $a \in \text{Im } \gamma$;
- (2) $\sum_a \dim E(S_a) = (1, 1, \dots, 1)$;
- (3) $\Omega^2(M) \in \mathcal{F}$ for any M in $\text{mod } A$.

Here, we denote by Ω^i the i -th syzygy of a module for any $i \in \mathbb{N}$.

Lemma 3.1. *Let M be an indecomposable module.*

- (1) [6, 3.2] *Either $\text{pd } M \leq 1$ or else $\text{top } \Omega^2(M) = \gamma(\text{top } M)$.*
- (2) [7, A.1] *The map $\gamma: \text{Im } \psi \rightarrow \text{Im } \gamma$ is bijective with inverse ψ .*

Lemma 3.2. [5, 3.1] Let S_a and S_b be simple modules with $\text{pd } S_a \geq 1$. Then S_b is a subfactor of $\Omega^2(S_a)$ (with multiplicity 1) if and only if $\psi(b) = a$.

Proposition 3.3. Let S_a and S_b be simple modules.

- (1) If $a \in \text{Im } \gamma$, then S_b is a subfactor of $E(S_a)$ if and only if $\psi(b) = \psi(a)$.
- (2) If $\text{pd } S_a \geq 2$, then $\Omega^2(S_a) \cong E(S_{\gamma(a)})$.

Proof. (1) Let S_b be a subfactor of $E(S_a)$ and $b \neq a$. Since S_a is the unique composition factor of $E(S_a)$ belonging to $\text{Im } \gamma$, then $\text{id } S_b = 1$. The sequence

$$0 \rightarrow S_b \rightarrow I_b \rightarrow I_{\tau^{-1}b} \rightarrow 0$$

is exact. Since $\dim_k I_b = \dim_k I_{\tau^{-1}b} + 1$, then

$$\psi(b) = \tau^{-q_b}(b) = \tau^{-q_{\tau^{-1}b}}(\tau^{-1}b) = \psi(\tau^{-1}b).$$

If $\tau^{-1}(b) = a$, we are done. If $\tau^{-1}(b) \neq a$, then $S_{\tau^{-1}b}$ is a subfactor of $E(S_a)$. Since $\tau^{-1}b$ is closer to a than b , we repeat this process and obtain $\psi(b) = \psi(a)$. By Lemma 3.1(2)

$$\sum_{a \in \text{Im } \gamma} \dim_k E(S_a) \leq \sum_{a \in \text{Im } \gamma} |\psi^{-1}(\psi(a))| = \sum_{a' \in \text{Im } \psi} |\psi^{-1}(a')| = n.$$

Then the equality holds and $\dim_k E(S_a) = |\psi^{-1}(\psi(a))|$. Therefore, S_b is a subfactor of $E(S_a)$ if and only if $\psi(b) = \psi(a)$.

(2) By Lemma 3.1(1) the tops of $\Omega^2(S_a)$ and $E(S_{\gamma(a)})$ are isomorphic. Since $\text{pd } S_a \geq 2$, we infer that $a = \psi\gamma(a)$. By Lemma 3.2 a simple module S_b is a subfactor of $\Omega^2(S_a)$ if and only if $\psi(b) = \psi\gamma(a)$. Then $\Omega^2(S_a)$ and $E(S_{\gamma(a)})$ have the same dimension. They are isomorphic. $\square$

Repeatedly removing the leaves, we obtain a sequence

$$\text{mod } A \xleftarrow{\quad} \text{mod } \epsilon(A) \xleftarrow{\quad} \cdots \xleftarrow{\quad} \text{mod } \epsilon^r(A)$$

of adjoint pairs of exact functors such that $\epsilon^r(A)$ is selfinjective.

For any $0 \leq m \leq r$, let $\mathcal{F}^m$ be the image of $\text{mod } \epsilon^m(A)$ in $\text{mod } A$ and let $E^m(S_a)$ be the image of S_a for each a in $\text{Im } \gamma^m$. Then $\mathcal{F}^m$ is an exact subcategory of $\mathcal{F}^{m-1}$ which is closed under projective covers, and $\{E^m(S_a) \mid a \in \text{Im } \gamma^m\}$ is a complete set of simple objects in $\mathcal{F}^m$.

Similarly, we have the following.

- (1) $\text{top } E^m(S_a) \cong S_a$ for any $a \in \text{Im } \gamma^m$;
- (2) $\sum_a \dim E^m(S_a) = (1, 1, \dots, 1)$;
- (3) $\Omega_{\mathcal{F}^{m-1}}^2(M) \in \mathcal{F}^m$ for any M in $\mathcal{F}^{m-1}$.

Lemma 3.4. Let $m \in \mathbb{N}$ and M be an indecomposable module.

- (1) [6, 3.3] Either $\text{pd } M \leq 2m - 1$ or else $\text{top } \Omega^{2m}(M) = \gamma^m(\text{top } M)$.
- (2) [7, A.1] The map $\gamma^m: \text{Im } \psi^m \rightarrow \text{Im } \gamma^m$ is bijective with inverse ψ^m .

Proposition 3.5. Let $m \in \mathbb{N}$ and S_a, S_b be simple modules.

- (1) If $a \in \text{Im } \gamma^m$, then S_b is a subfactor of $E^m(S_a)$ if and only if $\psi^m(b) = \psi^m(a)$.
 (2) If $\text{pd } S_a \geq 2m$, then $\Omega^{2m}(S_a) \cong E^m(S_{\gamma^m(a)})$.

Proof. (1) Let S_b be a subfactor of $E^m(S_a)$. If $m = 1$, then $\psi(b) = \psi(a)$ by Proposition 3.3(1). If $m \geq 2$, then S_b is a subfactor of an $E^{m-1}(S_{a'})$ with $\psi(a') = \psi(a)$. By induction hypothesis, $\psi^{m-1}(b) = \psi^{m-1}(a')$. Then

$$\psi^m(b) = \psi\psi^{m-1}(b) = \psi\psi^{m-1}(a') = \psi\psi^{m-1}(a') = \psi^m(a).$$

By Lemma 3.4(2) we have

$$\sum_{a \in \text{Im } \gamma^m} \dim_k E^m(S_a) \leq \sum_{a \in \text{Im } \gamma^m} |\psi^{-m}(\psi^m(a))| = \sum_{a' \in \text{Im } \psi^m} |\psi^{-m}(a')| = n.$$

Then the equality holds and $\dim_k E^m(S_a) = |\psi^{-m}(\psi^m(a))|$. Therefore, S_b is a subfactor of $E^m(S_a)$ if and only if $\psi^m(b) = \psi^m(a)$.

(2) By Lemma 3.4(1) the tops of $\Omega^{2m}(S_a)$ and $E^m(S_{\gamma^m(a)})$ are isomorphic. For any $1 \leq k \leq m$, the operation $\Omega_{\mathcal{F}^{k-1}}^2$ sends a simple object in $\mathcal{F}^{k-1}$ to zero or a simple object in $\mathcal{F}^k$ by Proposition 3.3(2). Since every $\mathcal{F}^k$ is an exact subcategory of $\mathcal{F}^{k-1}$ which is closed under projective covers, then

$$\Omega^{2m}(S_a) \cong \Omega_{\mathcal{F}^{m-1}}^2 \cdots \Omega_{\mathcal{F}^1}^2 \Omega^2(S_a)$$

is a simple object in $\mathcal{F}^m$. Then $\Omega^{2m}(S_a)$ is isomorphic to $E^m(S_{\gamma^m(a)})$. $\square$

Let $m \in \mathbb{N}$. Following [7, 3.5] we get that $\text{pd } S_a \neq 1, 3, \dots, 2m-1$ if and only if $a \in \text{Im } \psi^m$.

Proposition 3.6. Let $m \in \mathbb{N}$ and S_a be a simple module with $\text{pd } S_a \geq 2m-1$. Then S_b is a subfactor of $\Omega^{2m}(S_a)$ (with multiplicity 1) if and only if $\psi^m(b) = a$.

Proof. If $\text{pd } S_a = 2m-1$, then $a \notin \text{Im } \psi^m$ and $\Omega^{2m}(S_a) = 0$. If $\text{pd } S_a \geq 2m$, then $a \in \text{Im } \psi^m$ and $\psi^m\gamma^m(a) = a$. Since $\Omega^{2m}(S_a)$ and $E^m(S_{\gamma^m(a)})$ are isomorphic by Proposition 3.5, then S_b is a subfactor of $\Omega^{2m}(S_a)$ if and only if $\psi^m(b) = a$. $\square$

Theorem 3.7. Let A be a Nakayama algebra, and S_a be a simple module. Suppose the minimal projective resolution of S_a is periodic after $2r$. Then

$$\sum_{i=0}^{p-1} \dim_k \Omega^{2r+2i}(S_a) = n_a^*,$$

where p is the periodicity of R_A^* and n_a^* is the cardinality of the component containing a in R_A^* .

We mention that Theorem 2.6 follows by a dual version of this theorem.

Proof. By Proposition 3.6 a simple module S_b is a subfactor of $\Omega^{2r+2i}(S_a)$ if and only if $\psi^{r+i}(b) = a$. Then

$$\sum_{i=0}^{p-1} \dim_k \Omega^{2r+2i}(S_a) = \sum_{i=0}^{p-1} |\psi^{-r-i}(a)| = n_a^*.$$

The proof is finished. $\square$

4. Graded Cartan determinant

We calculate the determinant of the Cartan matrix of a Nakayama algebra with grading in this section.

Let A be a Nakayama algebra of order n . Suppose A is not selfinjective. Then the resolution quiver R_A has a leaf. We may assume n is a leaf. Let $e' = 1 - e_n$. The algebra $A' = e'Ae'$ is a Nakayama algebra of order $n - 1$.

For any grading $\mathbf{d}$ is on A , let $\mathbf{d}'$ be the grading on A' such that

$$\mathbf{d}'(x_a) = \begin{cases} \mathbf{d}(x_a) + \mathbf{d}(x_{a+1}) & \text{if } a = n - 1 \\ \mathbf{d}(x_a) & \text{otherwise.} \end{cases}$$

Lemma 4.1 ([3, 3.8]). *Suppose n is a leaf in R_A .*

- (1) $R_{A'}$ equals R_A with n removed;
- (2) $R_{A'}$ and R_A have the same weight;
- (3) $R_{A'}$ and R_A have the same number of components.

Lemma 4.2. *We have $\det C_A^{\mathbf{d}}(t) = \det C_{A'}^{\mathbf{d}'}(t)$.*

Proof. Since n is a leaf, the injective dimension of S_n is equal to 1. There is an exact sequence of graded modules

$$0 \rightarrow S_n \rightarrow I_n \xrightarrow{x_{n-1}} I_{n-1}(\mathbf{d}(x_{n-1})) \rightarrow 0.$$

Denote by r_a the a -th row of $C_A^{\mathbf{d}}(t)$. It is the graded dimension vector of the injective module I_a . The above sequence shows that

$$r_n - t^{\mathbf{d}(x_{n-1})} r_{n-1} = \epsilon_n.$$

Adding the product of $-t^{\mathbf{d}(x_{n-1})}$ times of r_{n-1} to r_n yields an elementary transformation

$$C_A^{\mathbf{d}}(t) \rightarrow \begin{pmatrix} C_1(t) & * \\ 0 & 1 \end{pmatrix}.$$

We have $\det C_A^{\mathbf{d}}(t) = \det C_1(t)$. It is routine to check that $C_1(t)$ is just the graded Cartan matrix of A' with grading $\mathbf{d}'$. $\square$

Let R_A be the resolution quiver of A . Denote by c the number of components in R_A and by w the weight of R_A . We denote by (a, b) the greatest common divisor of two integers a and b .

Lemma 4.3. *Let $A = k\Delta_n / \text{rad}^\ell$ be a selfinjective Nakayama algebra. Let $\mathbf{d}$ be a grading on A and d be the sum of the degrees of all arrows. Then*

- (1) $\det C_A^{\mathbf{d}}(t) = (1 - t^{d\ell/(n, \ell)})^{(n, \ell)} / (1 - t^d)$;
- (2) $w = \ell/(n, \ell)$ and $c = (n, \ell)$.

Proof. (1) Let D be the $n \times n$ matrix

$$D = \begin{pmatrix} 0 & 0 & \cdots & 0 & t^{\mathbf{d}(x_n)} \\ t^{\mathbf{d}(x_1)} & 0 & \cdots & 0 & 0 \\ 0 & t^{\mathbf{d}(x_2)} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & t^{\mathbf{d}(x_{n-1})} & 0 \end{pmatrix}$$

and let $\mathbb{I}$ be the $n \times n$ identity matrix. Then

$$C_A^{\mathbf{d}}(t) = \mathbb{I} + D + \cdots + D^{\ell-1} = (\mathbb{I} - D^{\ell})(\mathbb{I} - D)^{-1}.$$

Note that $\det(\mathbb{I} - D^k) = (1 - t^{dk/(n,k)})^{(n,k)}$ for any $k \in \mathbb{N}$. It follows that

$$\det C_A^{\mathbf{d}}(t) = (1 - t^{d\ell/(n,\ell)})^{(n,\ell)} / (1 - t^d).$$

(2) The admissible sequence of A is $(\ell, \ell, \dots, \ell)$. Then R_A consists of (n, ℓ) cycles of periodicity $n/(n, \ell)$. We have $w = \ell/(n, \ell)$ and $c = (n, \ell)$. $\square$

Theorem 4.4. *Let A be a Nakayama algebra with grading $\mathbf{d}$. Then the determinant of the graded Cartan matrix $C_A^{\mathbf{d}}(t)$ is*

$$\det C_A^{\mathbf{d}}(t) = (1 - t^{dw})^c / (1 - t^d),$$

where d is the sum of the degrees of all arrows.

Proof. If A is selfinjective, then the conclusion follows from Lemma 4.3.

Suppose A is not selfinjective. By Lemma 4.2 there is a Nakayama algebra A' with grading $\mathbf{d}'$ such that the determinants of $C_A^{\mathbf{d}}$ and $C_{A'}^{\mathbf{d}'}$ are equal.

By Lemma 4.1 R_A and $R_{A'}$ have the same weight and the number of cycles. Note that $\mathbf{d}'$ and $\mathbf{d}$ have the same total degree. Repeat this process, we get a selfinjective Nakayama algebra B with grading $\mathbf{f}$. Then

$$\det C_A^{\mathbf{d}}(t) = \det C_B^{\mathbf{f}}(t) = (1 - t^{dw})^c / (1 - t^d).$$

This finishes the proof. $\square$

Corollary 4.5. *Let A be a Nakayama algebra with length grading. Then the determinant of the graded Cartan matrix $C_A(t)$ is*

$$\det C_A(t) = (1 - t^{nw})^c / (1 - t^n),$$

where n is the order of A .

Declaration of competing interest

All authors disclosed no relevant relationships.

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